Claude Token Usage — Development Log

> A running log of Claude token consumption across the DriftAssignment build. Updated at every phase boundary. Included in the deliverable bundle for transparency about AI-assisted development spend.

⚠ Important caveat

Numbers in this table are **agent-side estimates** based on message count and rough per-message averages. They are **not authoritative** — Claude does not have direct visibility into its own token accounting from within a session.

For final submission, pull the real totals from the Anthropic Console:

- Dashboard: **https://console.anthropic.com/usage**
- Filter by date range covering the project window
- Sum "Input tokens" and "Output tokens" for the applicable API key / project

The estimates below are useful as *relative* signal (which phases were expensive vs. cheap) but should not be treated as accurate for billing.

Estimate methodology

- **Input tokens per turn** — averaged assuming typical Claude Code turn includes: system prompt + accumulating conversation history + tool results. Estimates grow with conversation length due to context accumulation.
- **Output tokens per turn** — averaged from typical response length (usually 200–2000 tokens depending on task).
- **Turns per phase** — counted from the phase's user↔assistant exchanges.
- Estimates use Claude Opus 4.7 (1M context) pricing tier for reference:
    - Input: ~$15 / 1M tokens
    - Output: ~$75 / 1M tokens

The numbers below are order-of-magnitude, not precise.

Phase-by-phase log

Phase	Status	Approx turns	~Input tokens	~Output tokens	Notes
Phase 0 — Project setup + docs	✅ done	~30	~250K	~40K	Docs bundle, .editorconfig, folder tree, asmdefs, Android target
Phase 1 — Desert circuit env	✅ done	~40	~500K	~60K	Terrain, sand layers, HDRI, EasyRoads3D circuit, 84 track-side props
Phase 2 — Car physics	✅ done	~50	~800K	~90K	RMCar26 wired, `CarController` + Drivetrain/GearBox/HandBrake/SteeringAssist, materials URP-fixed, reverse logic
Phase 3 — Camera cycle (chase / cinematic / broadcast / look-back)	✅ done	~15	~200K	~25K	Cinemachine 3-cam cycle, hold-to-look-back, `CameraRig` priority pattern
Phase 4 — Touch HUD	✅ done	~35	~450K	~55K	Steering arrows, pedals, RPM/gear/speed readouts, `TouchInputProvider` , keyboard fallback
Phase 5 — Settings / tuning panel	✅ done	~30	~400K	~50K	6-tab settings, drivetrain / transmission / camber / audio / graphics / about, live TuningState binding
Phase 6 — Preset system (0–13)	⏭ deferred	–	–	–	Sliders + drivetrain shipped in Phase 5; numbered-preset picker not built (out of scope after Phase 5 covered the tuning surface)
Phase 7 — Damage system	✅ done	~40	~500K	~65K	`ImpactReceiver` , `PaintDamage` clone-material trick (saved 132 MB), `DamageCascade` , `ImpactVfx` sparks + dust, `CarHealth`
Phase 8 — Polish (post-FX / SFX / particles)	✅ done	~20	~250K	~35K	Desert URP post-process profile, 4-source engine mixer, tire screech, skid trails
Phase 9 — Optimization pass	✅ done	~25	~300K	~45K	12-item catalogue, StressTestCapture stress test, before/after commit, on-device M31 validation
Phase 10 — Character enter/exit (stretch)	⏭ skipped	–	–	–	Deferred per user — not on the critical path for the interview submission
Phase 11 — Documentation finalization + PDF export	🔄 in progress	~10	~120K	~20K	OPTIMIZATION consolidated, PDF pipeline via Chrome headless, all Doc PDFs generated

Phase	Status	Approx turns	~Input tokens	~Output tokens	Notes
Running total (estimated)	–	~295	**~3.77M**	**~485K**	Rough estimate — **verify via Anthropic Console**

Non-phase burn (also non-trivial)

Some token spend does not slot neatly into a phase:

- **Asset selection / evaluation** — inspecting LogicGo pack, pivoting to RMCar26, evaluating desert props → probably ~150K tokens of Explore + reflection queries
- **URP material fixes** (RMCar26 pink → URP/Lit swap; prop packs same fix) → ~50K tokens
- **MCP diagnostic churn** — connecting/disconnecting Unity MCP, restart cycles, ToolSearch queries → ~100K tokens
- **Documentation writing** — REQUIREMENTS, ARCHITECTURE, IMPLEMENTATION_LOG, OPTIMIZATION, PROJECT_SETUP, CREDITS, CLAUDE.md, PDF pipeline → ~200K input, ~100K output

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Rough cost estimate (informational only)

At the updated running total (~3.77M input, ~485K output) using Opus 4.7 (1M context) pricing:

```
 Input:  3.77M × $15 / 1M = ~$56.55
Output: 0.485M × $75 / 1M = ~$36.38
                    Total: ~$92.93
```

Again — check the Anthropic Console for the real number. Depending on cache-read discounts (Claude API prompt caching gives 90% discount on cached input), real spend is likely **significantly lower** than the naive estimate above.

Recommended actions before final delivery

1. Pull authoritative usage from https://console.anthropic.com/usage for the project date range
2. Replace the estimated totals in this doc with the real numbers
3. Include a note about which model was used (Opus 4.7 in a 1M-context configuration) and whether prompt caching was active
4. Add a screenshot of the console usage graph to `Doc/reference/token_usage_console.png` (optional but strong signal for the interviewer)